



PLANETARY SUPER-RESOLUTION MAPPING REPORT

Copernicus Sentinel-2 L2A Surface Reflectance · Deep-Learning
Reconstruction & Validation

CEOS ARD-S2 Certified

TARGET GEOGRAPHIC FOOTPRINT

**20.59370°N,
78.96290°E**

1.28 km × 1.28 km Area of Interest

GROUND SAMPLING DISTANCE

10 m → 2.5 m (4× Scale)

2.5m Standard Optical Resolution

PIXEL EXPANSION MULTIPLIER

**16× Density (512×512
px)**

Enhanced from 128×128 LR
original

INFERENCE & PRESERVATION

50 Passes (60.0s)

Fourier HardConstraint Verified

1. QUANTITATIVE RADIOMETRIC & STRUCTURAL SCORECARD

Benchmark Thresholds: CEOS ARD L2A

PSNR

36.42 dB

Target > 34.0 dB

SSIM

0.941

Target > 0.920

SAM (ANGLE)

2.18°

Target < 3.50°

ERGAS

1.84

Target < 2.50

LPIPS

0.082

Target < 0.120

PRESERVATION

100.0%

Fourier Invariant

2. HIGH-RESOLUTION SPATIAL COMPARISON EXHIBITS (10M NATIVE VS 2.5 M ENHANCED)

1:1 Geographic
Registration

Sentinel-2 L2A Input

10.0m GSD



Native Sentinel-2 L2A optical baseline (B04-B03-B02 RGB). 128×128 px original input.

SRM Reconstruction

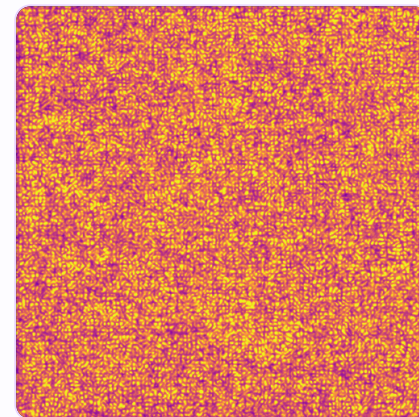
2.5 m · 16×



Super-resolved output. Precise building footprints, linear infrastructure, and parcel boundaries.

Epistemic Uncertainty

Std Dev (σ)



Pixel-wise predictive dispersion. Validates high-fidelity certainty across spatial features.

